DATABRICKS FDE BUILD, DEMO & DELIVERY

# ReelInsights Decision Support

Evolving the theater sales bundle from passive reporting to governed, closed-loop action

CURRENT-STATE DIAGNOSIS

# As-Is Scaffold & Hidden Gaps

`seed_theater_data`
**POS, Layout & IMDB**

→

`denormalized_theater_sales`
**Naïve Inner Join Drops Rows**

→

`Lakeview & App`
**Past KPIs & Bare App**

### Silent Data Loss

Orphan sales vanish unlogged.

### Unscoped Access

Zero role-level row filtering.

### Passive Metrics

No screen reallocation guidance.

### Read-Only UI

Actions spill into spreadsheets.

TARGET LAKEHOUSE ARCHITECTURE

# Medallion & Quarantine Engine

01 • BRONZE

**Raw Ingestion**

Existing DAB sources stay intact.

02 • SILVER

**Silver Quarantine**

Isolates corrupt POS records.

03 • GOLD

**Gold Serving**

Role-scoped metrics plus demand.

04 • ACTION

**Databricks App**

Closed-loop Delta write-back.

**<2%**

Screen occupancy driving losses

**1,291**

Null screen IDs quarantined

**0**

Orphan records silently lost

**100%**

Bad prices & duplicates isolated

GOVERNANCE & INTERACTIVE SERVING

# Unity Catalog RLS & App

## Row-Level Governance

**Theater Managers:** Scoped strictly to local theater data.

**Zonal Leaders:** Regional rollups across assigned districts.

**Executive Leadership:** Global rollups plus chain-wide outliers.

**DQ Dashboard:** Tracks quarantined POS errors live.

## Extended reel-insights App

**7-Day Demand Trends:** Surfaces trailing velocity and occupancy.

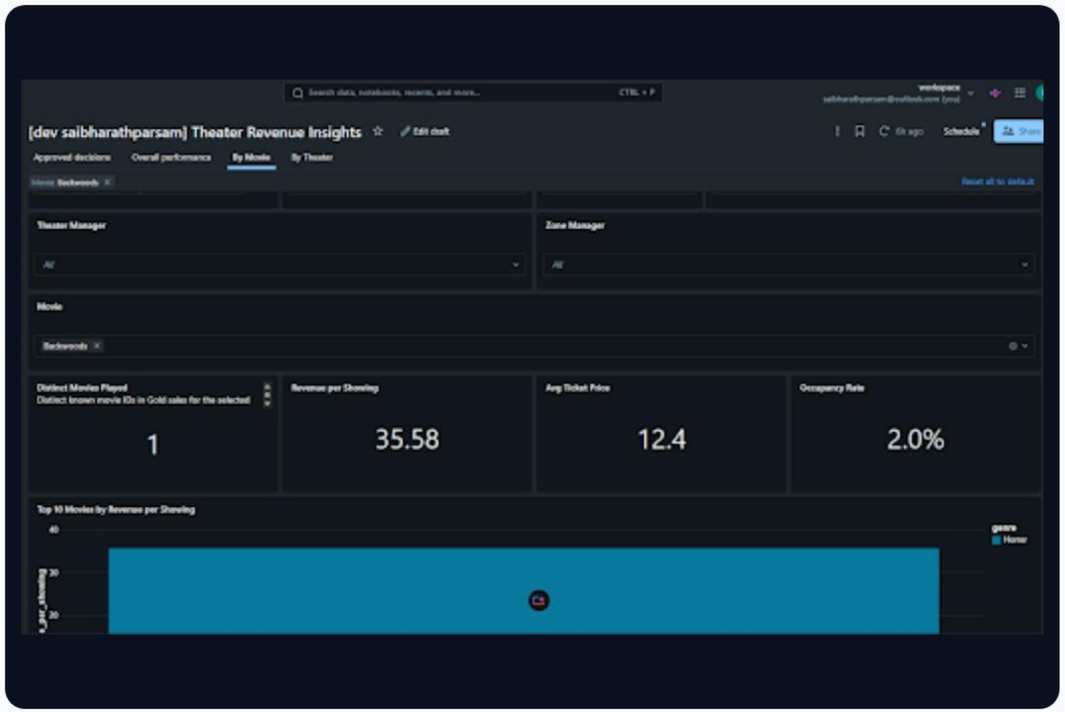
**Screen Reallocation:** Add or reduce screens by demand.

**In-App Approvals:** Replaces email and manual spreadsheets.

**Delta Write-Back:** Publishes decisions to `manager_feedback`.

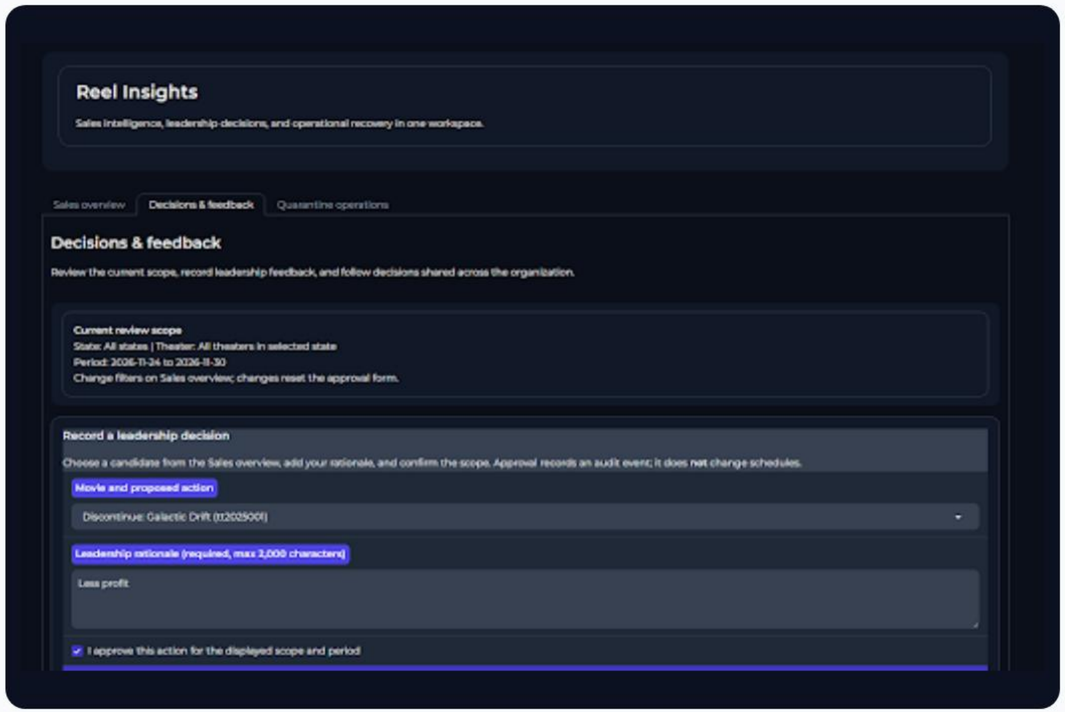
LIVE WORKING PROOF OF CONCEPT

# Working Dashboard & Databricks App




## Theater Revenue Insights Dashboard

Surfaces 2.0% occupancy rate by movie and zone.




## Reel Insights Databricks App

Logs leadership screen decisions directly to Delta.

FINANCIAL FEASIBILITY & DECISION

# MVP Cost, ROI & Pilot Approval

ESTIMATED MONTHLY MVP SPEND

## $600–$1,500

Serverless SQL, daily jobs, Medium App & 25% buffer.

MONTHLY BREAK-EVEN REVENUE

## $5,000/mo

Incremental box-office sales at 20% margin to cover $1k spend.

### Cost Controls

10–min auto-stop plus incremental MERGE.

### Databricks Stack

Unity Catalog, Delta, Apps & DAB.

### Approve Phase 1

Launch controlled pilot & RLS rollout.